A

GWAS SNP−Level Association

B

MAGMA Gene−Level Association

$-\log_{10}(p)$

15
10
5
0

1 2 3 4 5 6 7 8 9 10

Chromosome

C

CATFISH Pathway Enrichment

gluconeogenesis I [56]
L−glutamate biosynthesis IV [6]
L−glutamine degradation I [34]
glyoxylate cycle [17]
L−threonine biosynthesis [3]
phospholipases [106]
superpathway of phospholipid biosynthesis II (plants) [4]
CMP−3−deoxy−D−<I>manno</I>−octulosonate biosynthesis [3]
glutamate removal from folates [3]
folate polyglutamylation II [5]
trans−zeatin biosynthesis [21]
cis−zeatin biosynthesis [4]
choline biosynthesis I [12]
L−phenylalanine biosynthesis II [6]
aerobic respiration I (cytochrome c) [298]
β−alanine biosynthesis I [10]
2−oxoisovalerate decarboxylation to isobutanoyl−CoA [9]
very long chain fatty acid biosynthesis I [65]

ACAT
Fisher
Stouffer
minP
TFisher
Omnibus

$-\log_{10}(p)$
15
10
5
0